Rushil Rawat

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Education

University of Massachusetts, Amherst B.S. Computer Science + Business
Jr. CTO @ UMass Student Government · AI @ MIT · CodePath · Dean's List *Aug 2024 – May 2028*

Experience

IT Infrastructure & Service Operations Intern June 2026 – Aug 2026
Teachers' Retirement System of the City of New York *New York, NY*

- Built a **PowerShell** script to audit Active Directory groups on last-logon data, flagging **300+** stale accounts
- Supported **AWS EKS/Kubernetes** staging deployment checks using CodePipeline, CodeBuild, and CloudWatch
- Assisted with **AppLocker** allowlisting on 9 workstations and **BeyondTrust** access controls across 3 service tiers
- Triaged **12+ ServiceNow** tickets diagnosing Windows software, driver, connectivity failures via Remote Desktop

Full-Stack Developer Feb 2025 – Present
College of Humanities and Fine Arts, UMass Amherst *Amherst, MA*

- Automated migration of 12 course modules with a **Python Web scraper** and QA checks, achieving 0 data loss
- Diagnosed Canvas–Pressbooks sync failures and optimized **MySQL** queries, restoring access for 400+ students
- Developed **120+ interactive** HTML/CSS/JavaScript assessments with consistent cross-browser compatibility

Open Source Developer Jan 2025 – Apr 2025
CICS Open Source Apprenticeship, UMass Amherst *Hybrid*

- Prototyped **Rust** map-navigation features for Veloren (10,000+ commit), filing a waypoint bug with maintainers
- Configured **GitLab CI pipelines** to build and deploy team documentation for 3-person **Agile** Team
- Traced 8+ pull requests through review, merge to map **code-review** standards in open-source codebase

Projects

TableFinder @ UMass Amherst Dining | *React, TypeScript, Flask, Firebase* Sep 2025 – Present

- Built a dining hall occupancy platform with NFC check-ins, syncing 1,000+ real time table vacancy via **Firebase**
- Implemented **TTL-based session expiry** to auto-release stale table claims, removing manual checkout issues
- Validated with **100+** student testers under live dining-hall load, resolving check-in failures ahead of rollout

CapitolWatch Insider Trading | *Python, FastAPI, PostgreSQL, React, Docker* Jul 2026

- Designed **PostgreSQL** ETL pipeline for 72,000+ congressional trades using fuzzy name matching
- Evaluated trade timing of 350+ members with permutation testing and FDR correction, flagging 2 anomalies
- Reduced **ETL** pipeline time by 88% (4 hours to 29 mins) by content-hash **caching**, served via **FastAPI** and React

Sanskritized Hindi Lexicon | *React, TypeScript, CSS, JSON, NoSQL* May 2026

- Launched a dictionary reaching 1,000+ users, script-aware fuzzy search across 1800+ entries
- Cut load time with **lazy-loading**, paired debounced input with client-side indexing keeping search **NoSQL**
- Shipped audio pronunciation using **Web Speech API**, saving preferences locally with XSS-safe rendering

Skills

Languages: Python, Java, C++, TypeScript, JavaScript, SQL, Rust, PowerShell
Web & Backend: React, FastAPI, Flask, Node.js, REST APIs, Vite, Tailwind CSS, NumPy, scikit-learn
Databases & Infrastructure: PostgreSQL, MySQL, MongoDB, Firebase, AWS, Azure, Kubernetes, Docker
Tools & Testing: Git, Linux/Unix, CI/CD, pytest, Vitest, ServiceNow, Jira